MojoBrain

Architecture Guide

How to Build a 7-Agent AI Operating System

Mission Control — your live agent dashboard

eyshtech.vip | By Eysha Penrod

The Problem

You built a business. Now it's running you.

Every week you spend 15–20 hours on operations that have nothing to do with serving clients — follow-ups you forgot to send, invoices you haven't checked, that security setting you've been meaning to audit, the lead that replied three days ago and is now cold. You know the list. It haunts you on Sunday nights.

I know because I was you. I spent 6 years as a Google Workspace Sr. Account Manager at MarketStar, managing environments for 250+ SMBs. I watched the same pattern everywhere: smart business owners drowning in admin work that should run itself.

When I went out on my own, it got worse. I was the CEO, the IT department, the sales team, and the compliance officer — all at once. I almost hired a VA. Then I realized I'd just be managing the VA.

I remember the exact week it broke me. I had three client proposals sitting unfinished, a lead from Tuesday I hadn't followed up on, and I'd just discovered my SSL cert had expired three weeks ago and nobody told me. Not a tool. Not an alert. Nobody. Because there was nobody — it was just me.

That was the week I stopped trying to work harder and started building something to work for me.

> 15-20 hours/week on ops x $100/hr opportunity cost
> = $6,000-8,000/month in lost revenue.
> MojoBrain costs ~$5-20/mo in API fees to run.

So I built MojoBrain. Seven autonomous AI agents that handle lead gen, outreach, compliance, finances, and security. Running 24/7 on a single Mac Mini. I approve from my phone every morning. The agents handle the rest.

I named it Mojo. It felt right.

What Actually Runs

Let me walk you through it. MojoBrain is a collection of Python scripts that run on cron schedules — every 5 minutes, every 15 minutes, every hour, depending on the agent. Each one writes to a JSON state file. No fancy framework. No Docker. Just Python, cron, and JSON.

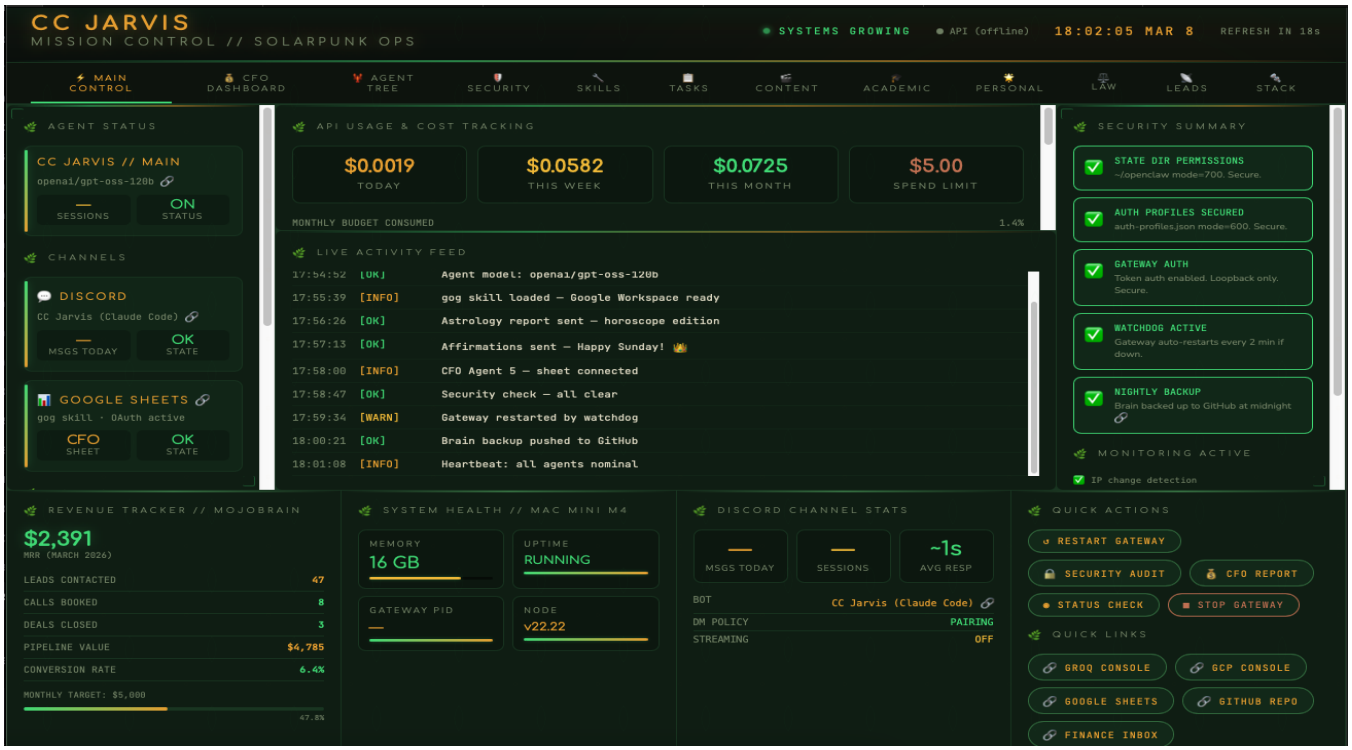
Here's the thing about how they coordinate — none of them talk to each other directly. Everything goes through the Nerve Center. Every 5 minutes, the Nerve Center reads all agent outputs, builds a unified picture of what's happening across the entire system, and routes events between agents when something cross-functional happens. If the Lead Engine finds a hot prospect and the CSO notices that prospect's domain has a security issue — the Nerve Center connects those dots.

You get notifications via Discord. Critical stuff goes to one channel. Routine logs go to another. Morning briefings go to a third. You see what matters. Nothing more.

> By the time I wake up, the CSO has already run 8 security checks,
> the Lead Engine has scored overnight scrapes, and my morning
> briefing is waiting in Discord. I read it over coffee. Most days
> I approve two things and move on with my actual work.

Here's what I don't do anymore: I don't manually check if my SSL cert is valid. I don't search LinkedIn for prospects. I don't track which leads I've emailed and which I haven't. I don't check my bank statements for weird charges. I don't scan for GDPR compliance gaps. I don't monitor disk space. The agents do all of it, every day, whether I'm awake or not.

MISSION CONTROL — LIVE VIEW OF ALL AGENTS

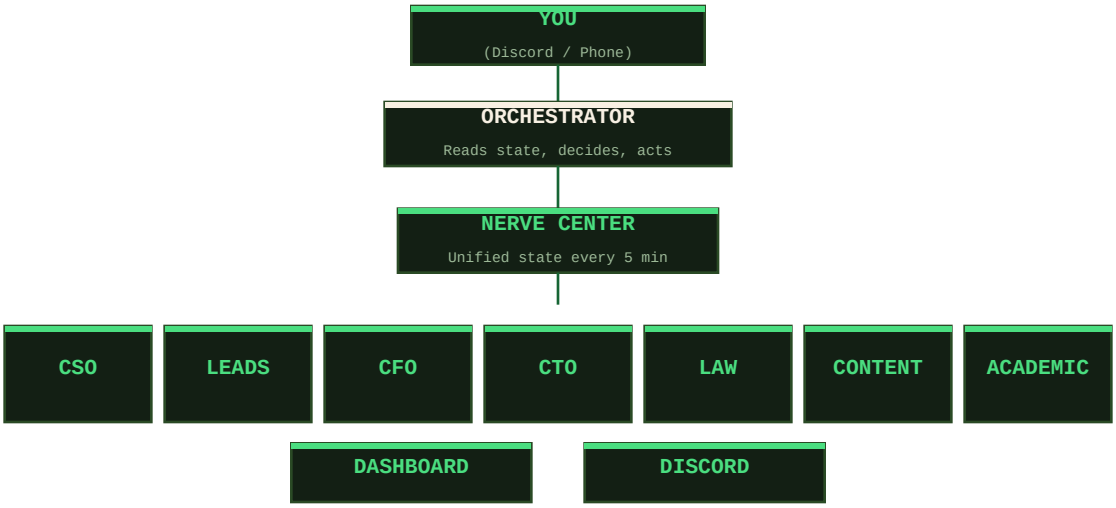


Real-time agent status, financial tracking, security alerts. One screen.

How It All Connects

The architecture is deliberately simple. Complexity is the enemy of reliability — and when you're running a business on this system, reliability isn't optional.

SYSTEM ARCHITECTURE



You sit at the top. The Orchestrator reads the Nerve Center's unified state and decides what needs your attention. The Nerve Center reads all 7 agent outputs every 5 minutes. The agents run independently on their own cron schedules, writing to JSON state files. Everything flows down to the Dashboard (your visual command center) and Discord (your notification channel).

There's no single point of failure below the Orchestrator level. If one agent crashes, the others keep running. The Nerve Center notices the gap and flags it. Most of the time, by the time you see the alert, the self-heal agent has already restarted whatever broke.

The 7 Core Agents

Think of them less like software and more like employees who never sleep, never forget, and never complain. Each one has a job, a schedule, and a personality. Here's who does what.

Orchestrator

CENTRAL INTELLIGENCE

The brain. Reads all agent outputs, makes decisions, routes events. It's the one that sends you your morning briefing and decides what deserves your attention and what can wait. Every morning at 7am, it synthesizes overnight activity into a 3-paragraph summary: what happened, what needs your attention, and what it handled without you. Most days, the third category is the longest.

CSO

SECURITY

Paranoid in the best way. Checks DNS records, scans for breach mentions, audits SSL certs — every 15 minutes, without being asked. I haven't thought about email security since I deployed it. Last month it caught a misconfigured SPF record that would have sent all my outreach emails to spam. I didn't even know it was broken. The CSO did.

Lead Engine

LEAD GENERATION

This is the one that paid for itself first. Scrapes, validates, scores, and queues leads while you sleep. Replaced 15 hours of manual prospecting per week. Last Tuesday it found a design agency that had just posted three Workspace admin jobs. By 6am it had validated their domain, found the ops director's email, scored them 8/10, and queued a personalized outreach. I didn't touch any of it.

Nerve Center

COORDINATION

The nervous system. Reads ALL agent outputs every 5 minutes, builds a unified state, detects cross-agent patterns. This is what makes it a system, not just scripts. When the Lead Engine finds a prospect and the CSO notices their domain has issues — the Nerve Center connects those dots and routes an alert. It's the difference between 7 independent tools and one coordinated brain.

CFO

FINANCE

Tracks your burn rate, monitors subscriptions, flags suspicious charges, knows your tax deadlines. Your financial watchdog that never takes a day off. It caught a $49/mo subscription I'd forgotten about — some analytics tool I signed up for during a free trial six months ago. That's $294 I would have kept bleeding without noticing.

CTO

INFRASTRUCTURE

Monitors system health, disk space, cron jobs, and runs automated recovery when things go sideways. Because things always go sideways at 3am. It once restarted a crashed process, verified the fix, and sent me a Discord message — all before my morning alarm. I woke up to “Issue detected and resolved” instead of “Everything is down.”

Law

COMPLIANCE

GDPR scanning, disclaimer tracking, contract clause library, compliance routing. Keeps you covered without a $500/hr retainer. It runs weekly audits against your public-facing pages and flags anything that’s drifted out of compliance. Most businesses don’t even check this stuff until they get a letter. The Law agent checks it every 7 days.

What This Costs to Run

Here's exactly what this costs to run. I was shocked when I added it up the first time.

~$20	7	24/7
Monthly API Cost	Autonomous Agents	Unattended Operation

Component	Technology	Cost
Hardware	Mac Mini / Linux / VM	$0 (use yours)
LLM	Your choice (Claude, GPT, Groq)	$5-20/mo
Orchestration	Python 3 + cron	$0
Dashboard	Self-contained HTML	$0
Notifications	Discord webhooks	$0
Email	Brevo (5K emails/mo)	$9/mo
Remote Access	Tailscale	$0
Domain	Namecheap	$7/yr
Hosting	GitHub Pages	$0

The $0 cells aren't a trick. Groq gives you a genuinely useful free tier. Discord is free forever. GitHub is free for private repos. The only real cost is your LLM provider — and that's the brain that makes everything else work.

Why not just use Zapier?

I get this question a lot. Zapier, Make, n8n — they're all great tools. I've used them. But there's a fundamental philosophical difference between renting automation and owning your infrastructure.

Zapier is great until it isn't. The moment you hit a rate limit, or they change pricing, or they deprecate an integration you depend on, your entire business automation breaks. And you're stuck waiting for someone else to fix it. With MojoBrain, you own every line of code. If something breaks at 3am, you can fix it at 3am. Or better yet — your CTO agent fixes it for you.

The other thing nobody talks about: Zapier bills per task. When you're running 7 agents that each execute dozens of operations per day, those per-task costs add up fast. MojoBrain runs unlimited tasks for a flat API cost. The more your agents do, the more you save.

How I Got Here

MojoBrain didn’t start as a product. It started as survival.

I was running a consulting business while employed full-time, managing tools manually, and losing leads because I couldn’t follow up fast enough. I’d built this Frankenstein gateway thing — cobbled together from open-source tools, API proxies, and hope. It worked. Mostly.

Then it crashed at 3am on a Tuesday and took down everything. Not just the gateway — the crash triggered a cascade. The LLM proxy went down, which meant agent calls started failing, which meant the health checker started spamming restart attempts, which filled up the disk with error logs, which crashed the dashboard. I woke up to 178 error messages in my Discord.

That was the moment I decided to rebuild from scratch. Not fix it. Replace it.

> The first agent I built was the Lead Engine.
> Night one, it scraped and scored 47 leads while I slept.
> That one agent replaced 15 hours of manual prospecting per week.

The first morning after deploying the Lead Engine, I woke up to 23 scored leads waiting in Discord. I just sat there staring at my phone. It had done in one night what would have taken me two full days. And it ranked them — the ones most likely to buy were at the top. I’d never had that level of intelligence in my prospecting before.

Within two weeks I had 7 agents running. The system evolved from a collection of scripts into something that felt alive — a coordinated operating system with a central nervous system, a dashboard, and autonomous decision-making.

> The breakthrough wasn't any single agent.
> It was the Nerve Center — the component that reads ALL agent
> outputs every 5 minutes and coordinates cross-agent responses.
> That's when it stopped being scripts and became a system.

What I learned about reliability: simple beats clever every time. Cron jobs are boring. JSON files are boring. Python scripts that write to a file and exit are boring. But boring is what runs at 3am without crashing. The gateway crashed because it was clever. MojoBrain works because it’s boring.

I’m telling you this because if your system is currently broken, the answer is probably not to fix it. It’s to replace it with something simpler.

Start Here

The first agent I'd build if I were starting over is the security monitor. Not because it's the flashiest — it's not. But because it runs silently in the background and one day it's going to catch something that would have cost you real money or real embarrassment. Here's how to build all three.

01 — System Health Monitor

Before you start, you need: Python 3 installed, a Discord server with a webhook URL, and the psutil library (pip install psutil). That's it.

The script checks three things: disk space (alert if below 10%), memory usage (alert if above 90%), and whether your other cron jobs actually ran (check log file timestamps). It writes a status JSON file every run, and pings Discord only when something is wrong.

The key decision: how often to run it. I started with every hour. Within a week I moved it to every 15 minutes. When your server is your business, 45 minutes of undetected downtime is 45 minutes too many.

Watch out for false positives. macOS reports disk usage differently than Linux, and if you're running Time Machine backups, your disk will look "full" when it isn't. Add exclusions for known safe states.

Success looks like: you stop worrying about your infrastructure. You just trust that if something breaks, you'll know within 15 minutes. That peace of mind is worth more than the 30 minutes it takes to build.

```
Stack: Python + psutil + Discord webhook
  Schedule: */15 * * * *
  Build time: 30 minutes
```

02 — Morning Briefing Bot

You need: API access to whatever tools you use daily (email, calendar, task manager), plus a Discord webhook or email sender. The briefing bot pulls data from your tools and formats it into a human-readable summary.

Start simple. Mine started with three sections: email count, today's calendar, and overdue tasks. Over time I added weather, lead pipeline status, and financial alerts. Don't try to build the full version first — start with whatever makes your morning better and add from there.

The key insight: this agent changed my relationship with my phone. Instead of opening five apps and piecing together my day, I open Discord and it's all there. Formatted. Prioritized. Ready. It's like having a chief of staff who works for free and never calls in sick.

```
Stack: Python + cron + API calls
  Schedule: 0 11 * * * (7am ET)
  Build time: 1 hour
```

03 — Lead Scraper + Scorer

This is the one that makes money. Before you start, define your ideal customer profile — industry, company size, tech stack, geography. The more specific, the better the results. “Small businesses” is too broad. “Marketing agencies with 5-20 employees that use Google Workspace” is perfect.

The script searches DuckDuckGo (no API key needed) for businesses matching your criteria, validates their domains with DNS checks (kills parked/dead domains before you waste time), and scores them on signals like recent hiring posts, tech stack indicators, and company size. The output is a ranked JSON file that your morning briefing can read.

The decision that matters most: your scoring weights. I spent a week tuning mine. Hiring posts for IT roles are a strong signal. A company blog that hasn’t been updated in 2 years is a weak one. Don’t overthink it on day one — deploy, review the results, and adjust the weights based on which leads actually convert.

Watch out for rate limiting. DuckDuckGo will throttle you if you fire 200 searches in a minute. Add delays between requests. I run mine at 2am when there’s no competition for bandwidth.

```
Stack: Python + DuckDuckGo + DNS checks
  Schedule: 0 6 * * * (2am ET)
  Build time: 2-3 hours
```

Where to Go From Here

If you've read this far, you're probably one of two people.

Either you're going to build this yourself — great, the patterns are all here. The three agents above are a real starting point, not a teaser. Build them this weekend and you'll have a system that's already better than what 90% of businesses are running.

Or you're thinking "I want this but I don't want to build it."

A word on DIY vs done-for-you

If you're comfortable in a terminal and have a weekend to spend — build it. Seriously. The patterns are all here and the satisfaction of watching your first agent run autonomously is genuinely one of the best feelings in tech. You'll stay up until 2am not because you have to, but because you can't stop adding features.

If you're not technical, or you just want it running by next week — that's what the paid tiers are for. No judgment either way. Different people need different things. I've built MojoBrain for both kinds of people.

Three ways to work together:

Playbook — $297	All 7 agent scripts, Mission Control dashboard, deployment playbook with video walkthroughs. You deploy it yourself on your own hardware. You own it forever.
Guided Setup — $797	Everything in Playbook plus a 2-hour live setup call, your Lead Engine customized to your specific market, cold email templates, the full skills library, and 30 days of Discord DM support. Most people pick this one.
Managed Deploy — $1,497	I install everything on your machine, configure all 7 agents for your business, build 3 custom agents for your specific workflow, and support you for 90 days. You just show up and approve.

Not sure which one? Start with a $97 Workspace audit. Thirty minutes, actionable recommendations, and the $97 applies as credit if you go with any MojoBrain tier. It's the lowest-risk way to see if we're a good fit.

> Currently onboarding 3 clients per month.
> Book a free discovery call: calendar.app.google/35PXh7ZMU7BthPPV9

About

I'm Eysh. I spent 6 years as a Sr. Account Manager for Google Workspace at MarketStar — which means I spent 6 years watching small businesses struggle with the same operational chaos, over and over. The same missed follow-ups. The same security gaps. The same "I'll get to it next week" that turns into never.

When I went out on my own, I built the system I wish my clients had always had. MojoBrain runs in production daily, managing my own business operations. Seven agents, one machine, zero downtime. I sell the same system I depend on.

If something in this guide resonated — let's talk.

Website	eyshtech.vip
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Book a Call	calendar.app.google/35PXh7ZMU7BthPPV9
$97 Audit	buy.stripe.com/6oU14n2xY2kR5f06Xp6AM09
LinkedIn	linkedin.com/in/eysha-penrod